

Sec 7.2 – Integration by Parts

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Last section we learned a technique of integration, Substitution, that reversed the Chain Rule. Today, we introduce **Integration by Parts**, which reverses the Product Rule. Note that there will be questions on the Final Exam requiring Integration by Parts.

Let's start with the product of two functions of x , $u(x) \cdot v(x)$. For convenience we will just write this product as $u \cdot v$. If we differentiate this product, we use the Product Rule:

$$\frac{d}{dx} u \cdot v = u'v + uv'$$

If we integrate both sides of the equation above:

$$\int \frac{d}{dx} u \cdot v \, dx = \int u'v + \int uv'$$

The integral and derivative on the left-hand side cancel, which gives us:

$$u \cdot v = \int u'v + \int uv'$$

and rearranging:

$$\int uv' = u \cdot v - \int u'v$$

Let's try this on $\int x \cdot \sin x \, dx$. We need to choose a u and v . There is a bit of an art to it, we need to choose u and v so that the integral on the right-hand side is easier than the original integral on the left-hand side.

$$\int x \cdot \sin x \, dx$$

I always make it a table, like below, choosing u and v' and then determining what u' and v are.

- Whatever you choose for v' , you need to be able to find v .
- It helps if u' is simpler than u (or at least not more complicated).
- It helps if v' is simpler than v (or at least not more complicated).

Initially we have:

$u = x$	$v =$
$u' =$	$v' = \sin x$

and then it is easy to determine u' and v :

$u = x$	$v = -\cos x$
$u' = 1$	$v' = \sin x$

Plugging back in to the Integration by Parts formula gives:

$$\begin{aligned}\int x \cdot \sin x \, dx &= -x \cdot \cos x - \int \cos x \, dx \\ \int x \cdot \sin x \, dx &= -x \cdot \cos x + \int \cos x \, dx + C \\ \int x \cdot \sin x \, dx &= -x \cdot \cos x + \sin x + C\end{aligned}$$

Partner up

$$\int x e^x dx$$

$$\int_0^2 x e^x dx$$

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$$\int x e^x dx$$

Let $u = x$ and $v' = e^x$. Then $u' = 1$ and $v = e^x$ and

$$\int x e^x dx = x e^x - \int 1(e^x) dx = x e^x - e^x + C$$

$\int_0^2 x e^x dx$ We already evaluated this integral and can use the antiderivative from above:

$$\int_0^2 x e^x dx = (x e^x - e^x) \Big|_0^2 = (2e^2 - e^2) - (0e^0 - e^0) = e^2 + 1$$

Partner up

$$\int x \ln x dx$$

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$$\int x \ln x dx$$

Let $u = \ln x$ and $v' = x$. Then $u' = 1/x$ and $v = x^2/2$. (Choosing $v' = \ln x$ isn't a good choice since we don't have the antiderivative! Also, the derivative of $\ln x$ is much nicer!)

Then

$$\begin{aligned} \int x \ln x dx &= \ln x (x^2/2) - \int \left(\frac{1}{x}\right) \left(\frac{x^2}{2}\right) dx \\ &= \frac{x^2 \ln x}{2} - \int \frac{1}{2} x dx \\ &= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C \end{aligned}$$

Let's try another integral that looks like some of the assigned homework problems.

$$\int \sin^2 x \, dx = \int \sin x \cdot \sin x \, dx$$


 hint

$$u = \sin x \qquad v = -\cos x$$

$$u' = \cos x \qquad v' = \sin x$$

$$\begin{aligned} \int \sin x \cdot \sin x \, dx &= -\sin x \cdot \cos x - \int (\cos x) \cdot (-\cos x) \, dx \\ &= -\sin x \cdot \cos x + \int \cos^2 x \, dx \end{aligned}$$

This looks similar, so we will use a trigonometry substitution (note, any of these will be provided), $\cos^2 x = 1 - \sin^2 x$

$$= -\sin x \cdot \cos x + \int (1 - \sin^2 x) \, dx$$

$$\int \sin^2 x \, dx = -\sin x \cdot \cos x + \int 1 \, dx - \int \sin^2 x \, dx$$

But we have a $\int \sin^2 x \, dx$ on each side of the equation, so collect like terms!

$$2 \int \sin^2 x \, dx = -\sin x \cdot \cos x + \int 1 \, dx$$

$$2 \int \sin^2 x \, dx = -\sin x \cdot \cos x + x + C$$

and dividing by 2:

$$\int \sin^2 x \, dx = -\frac{1}{2} \sin x \cdot \cos x + \frac{x}{2} + C$$

We now have a technique to determine $\int \ln x \, dx$

$$u = \ln x \qquad v = x$$

$$u' = \frac{1}{x} \qquad v' = 1$$

$$\begin{aligned} \int \ln x \, dx &= x \cdot \ln x - \int \frac{1}{x} \cdot x \, dx \\ &= x \cdot \ln x - \int 1 \, dx \\ &= x \cdot \ln x - x + C \end{aligned}$$

There are more complex integrals discussed at the end of the section in the textbook.